

ADS corpus, 1970--2026 query

core tier 15,650 papers, exact-phrase query
cited tier 126,108 papers they reference

citation graph

380,361 directed arcs from ADS reference lists
direction collapsed, reciprocal pairs kept as parallel edges
13,800 nodes, 380,361 edges (multigraph)

consensus over $R = 30$ runs

Leiden, Reichardt and Bornholdt, $\gamma = 1$
single runs agree only at ARI 0.669 ± 0.083
co-association on existing edges, $\tau = 0.5$:
5 constructions agree at ARI 0.9294 (0.9102--0.9576)
14 communities of ≥ 30 , membership $Q = 0.4636$

canonical partition

validation of the community map

significance

DCM null, 50 draws,
best-of-5 Q both sides:
 $Q_5 = 0.4651$ vs 0.2652
 $\Delta Q = 0.1999$, $p = 1/51$

other algorithms

seeded, frozen graph:
Louvain 76.7% one-to-one
(75.8--81.9%, 5 schedules)
Infomap NMI 0.69, finer
(43 communities)

corpus cuts ($R = 10$)

5 cuts, 5 seed schedules,
25 matched deletions each:
no cut in the control tail
in every schedule (max 2/5)

representation

multigraph against weighted,
10 matched seeds: ARI 0.875
against 0.648 on reseed,
so it matters less than seed

temporal modularity

the same graph, layered by time
multiplex Q_T , layers of ΔT years
its communities span 20% to 28% more,
so the static partition is time-confined

findings

14 macrocommunities; the fixed dark-matter
set loses 91% of its members to the keyword
cut in all 5 schedules (co-location 0.0227--0.0341);
controls retain and co-locate ≥ 0.9432 :
an indexing artefact

arrows run from what is built to what is measured on it; the boxed group gathers four independent checks on the canonical partition